Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2021	ENGI3371-WE01

Title: **MEng/BEng:** Electrical Engineering 3		
Time Allowed:	2 hours	
Additional Material provided:	No	
Materials Permitted:		
Calculators Permitted:	Yes	Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: Yes		

Instructions to Candidates:	Answer ALL questions. All relevant workings must be shown.

Revision:	

Section A

Question 1

A small-scale direct drive three-phase wind turbine generator (WTG) system is used to provide energy to a local rural grid. The WTG has a three-phase permanent magnet synchronous generator which is interfaced to the local AC grid via a 4-quadrant fully rated back-to-back Voltage Sourced Converter (VSC) link in which the two converters share a common DC link.

At rated wind speed, the generator provides 0.8320 MW of active power to the grid at 0.98 lagging power factor and the machine-side converter sets a voltage of 690 V (line-line, RMS) at the machine terminals. The machine per-phase synchronous reactance at 50 Hz is 0.2 Ω.

All converters are loss-less.

Answer the following questions:

a) Sketch a block diagram of the WTG system with all components clearly labelled.

[10%]

b) Using the Single-Machine Infinite Bus (SMIB) model, calculate the magnitude and phase-angle of the per-phase armature (stator) current in the machine at rated operation. Meanwhile, using the same model, calculate the magnitude of the machine per-phase internal (no-load) EMF at rated operation and its load angle.

[40%]

c) Determine the total active and reactive power at the machine side converter and specify whether the machine is absorbing or injecting reactive power at its terminals. You should justify your answers in either case.

[20%]

d) Following a sudden gust of wind, the generator speed has increased by 25% however the generator active power has been kept at its rated value with turbine pitch regulation. Determine the new load angle in this case.

[30%]

continued

Question 2

Traction in an electric train is provided by a 4-quadrant (4Q) DC motor drive system. The drive system is fed by a three-phase 600 V (line-line, RMS), 50 Hz AC power supply through a 6-pulse (three-phase) Thyristor-bridge converter which shares a common DC link with the 4Q DC-DC machine-side chopper. The DC link voltage, which is stabilised by a capacitor is kept at 750 V (DC).

The DC motor has negligible armature resistance and at rated operation it rotates at 1500 RPM providing a full-load torque of 670 NM. The field in the DC motor is kept constant under normal operation. The motor constant is 0.39 V/RPM.

All converters are loss-less.

Answer the following questions:

a) Sketch the full circuit diagram of the entire traction system (DC motor drive and the grid-side converter) with all components clearly labelled.

[20%]

b) Assuming the motor has negligible mechanical losses, calculate the following values at rated operation (i.e., nominal speed and full-load torque).

(i) The duty ratio of the DC motor drive.

(ii) The total power delivered to the wheels of the train by the DC motor.

(iii) The armature current, drawn by the motor.

(iv) The current drawn by the 4Q DC chopper.

[40%]

c) Determine the firing angle of the 6-pulse Thyristor converter under rated operation.

[10%]

d) Under braking conditions, the motor brakes from its nominal speed to a stop whilst maintaining a braking torque of 80% of its full-load torque. Assume the motor maintains the same armature voltage as its rated operation during the initial stages of braking. Calculate the following for when motor is under braking conditions:

(i) The total braking power.

(ii) The armature current in the motor.

[30%]

continued

Section B

Question 3

Consider a 50Hz, three-phase transmission line with the following parameters per phase:

Series Resistance, $R = 10\,\Omega$

Series Reactance, $X = 50\,\Omega$

Shunt Admittance, $Y = 5 \times 10^{-4} \angle 90°\ S$

I. If the load at the receiving end of the line is 35 MW at 220 kV with a power factor of 0.85 lagging, find the magnitude and the phase angle of the sending end voltage. Take the receiving end voltage phase angle as the reference in this case.

[30%]

II. If the load at the receiving end is 35 MW with a power factor of 0.85 lagging, and the sending end voltage is 220 kV, find the magnitude of the receiving end voltage. Also find the difference between phase angles of the sending end and receiving end voltages.

[70%]

Question 4

a) In the two-bus system shown in Figure Q4, bus 1 is a slack bus with $V_1 = 1.0\angle 0°$ pu. A load of 80 MW and 20 MVAr is taken from bus 2. The line impedance is Z where $|Z| = 0.2083$ pu on a base of 100 MVA. Using the Newton-Raphson method, with an initial estimate of $|V_2|^{(0)} = 1.0$ pu and $\delta_2^{(0)} = 0°$ and for the first iteration, the diagonal elements of the Jacobian matrix are equal to 4. Obtain the voltage magnitude and phase angle of bus 2 after the first iteration.

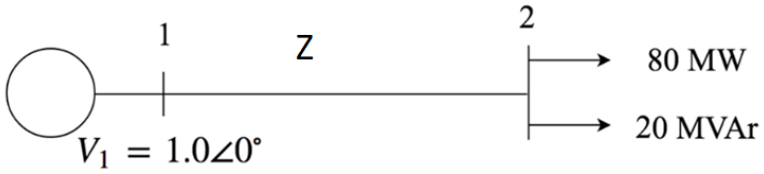


Figure Q4 network diagram

[40%]

b) Write an essay discussing how FACTS technologies can be used to facilitate increasing the share of renewable energy resources in the power system. Within the essay, first discuss the challenges of integrating renewable energies in the power system, including challenges in power system planning, control, optimisation, fault ride through, and power quality. Then, discuss how FACTS technologies can be used to address these challenges.

[60%]

END